

TEPL 1110 – Plumbing Apprentice IA (3 Credits)

Course Description

The Plumbing IA course explores the secure and highly demanded profession of plumbing. The primary emphases include: the understanding and interpretation of the International Plumbing Code (IPC), blueprint reading, materials and requirements, practical mathematics required in plumbing, and other subject areas that are essential to the trade which are more conducive to the classroom setting than a work environment.

Course Objectives

- Practice safety and the proper use of tools and leveling instruments.
- Define hydraulics and pneumatics.
- Interpret building and plumbing codes.
- Apply basic mathematics toward measurements, angles, slopes, and other plumbing related problems.
- Fabricate plumbing projects in a lab setting.

Course Outline

- Year IA

Textbook & Reading Materials

Mathematics for Plumbers and Pipefitters (ISBN: 9781111642600)

Modern Plumbing eBook Bundle w/2yr access code (ISBN: 9781637768303)

International Plumbing Code 2021 (ISBN: 9781609839628)

International Plumbing Code 2021 Study Companion (ISBN: 9781955052283)

Assignments & Assessments

On Campus

Orientation

Orientation Acknowledgement

Hybrid Assignments

Course Read and Review: Chapter 1 Safety Reading and Know and Understand Questions

Discussion 1: Safety

Workbook: Chapter 1 Lab Workbook

Modern Plumbing Chapter 1 Quiz

Read and Review: Chapter 2 Plumbing Tools Reading and Know and Understand Questions

Discussion 2: Plumbing Tools

Workbook: Chapter 2 Lab Workbook

Modern Plumbing Chapter 2 Quiz

Study Resources

Chapter 1 PowerPoint

Chapter 2 PowerPoint

Hybrid Assignments

Read and Review: Chapter 3 Plumbing Career Opportunities Know and Understand Questions

Discussion 3: Career Opportunities

Workbook: Chapter 3 Lab Workbook

Modern Plumbing Chapter 3 Quiz

Read and Review Chapter 4 History of Plumbing Know and Understand Questions

Discussion 4: History of Plumbing

Workbook: Chapter 4 Lab Workbook

Modern Plumbing Chapter 4 Quiz

Study Resources

Chapter 3 PowerPoint

Chapter 4 PowerPoint

Hybrid Assignments

Read and Review: Chapter 5 Leveling Instruments Know and Understand Questions

Discussion 5: Chapter 5

Workbook: Chapter 5 Lab Workbook

Modern Plumbing Chapter 5 Quiz

Read and Review: Chapter 7 Hydraulics and Pneumatics Know and Understand Questions

Discussion 6: Hydraulics and Pneumatics

Workbook: Chapter 7 Lab Workbook

Modern Plumbing Chapter 7 Quiz

Study Resources

Chapter 5 PowerPoint

Surveying Video 1

Surveying Video 2

Surveying Video 3

Surveying Video 4

Surveying Video 5

Chapter 7 PowerPoint

Hybrid Assignments

Read and Review: Chapter 8 Print Reading and Sketching Know and Understand Questions

Discussion 7: Identification of Plumbing Symbols

Workbook: Chapter 8 Lab Workbook

Modern Plumbing Chapter 8 Quiz

Read and Review: Chapter 9 Green Construction Know and Understand Questions

Discussion 8: What is Green Construction?

Workbook: Chapter 9 Lab Workbook

Modern Plumbing Chapter 9 Quiz

Study Resources

Chapter 8 PowerPoint

Chapter 9 PowerPoint

Hybrid Assignments

Read and Review: Chapter 10 Building and Plumbing Codes Know and Understand Questions

Discussion 9: Importance of Building Codes as They Apply to the Plumbing Industry

Workbook: Chapter 10 Lab Workbook

Modern Plumbing Chapter 10 Quiz

Understanding International Plumbing Code: Chapter 1 - Scope and Administration

Understanding International Plumbing Code: Drainage Pipe Sizing

Discussion 10: Building Code Part 1 and 2 videos

Study Resources

Chapter 10 PowerPoint

Hybrid Assignments

Read International Plumbing Code Chapter 1: Scope and Administration

IPC SS 1 - Scope and Administration

Discussion 10: Chapter 1 Scope and Administration

Read International Plumbing Code Chapter 2: Definitions

Discussion 11: Definitions?

On Campus Pass-off

Lab Activity Week 6: Copper Solder Project

Lab Activity Week 6: PVC project

Hybrid Assignments

Read International Plumbing Code Chapter 3: General Regulations

Discussion 13: Online Oass Feb 21st

301-306 Video

301-306 Video Quiz

IPC ss 3· General Regulations I
 Discussion: Chapter 3 International Plumbing Code
 Study Session 2: Article 301-307 from the International Plumbing Code
 Study Resources
 Hybrid Assignments
 CLASS DISCUSSION BDARD, WORK DN MISSING
 WORK AND OTHER ASSIGNMENTS
 Study Session: Article 308-314 from the International Plumbing Code
 307-316 Video
 IPC 307-316 Video Quiz
 IPC 55 3 - General Regulations II
 Read and Review Chapter 6 Mathematics for Plumbers Know and Understand Questions
 Discussion: What math topics are you most concerned about?
 Workbook: Chapter 6 Lab Workbook
 Modern Plumbing Chapter 6 Quiz
 On Campus Pass-off
 Instructor Demonstration: Soldering Demonstration
 Study Resources
 Chapter 6 PowerPoint
 Hybrid Assignments
 Read Mathematics for Plumbers and Pipefitters Unit 1: Basic Rules of Math.
 MPPUnit1
 Read Mathematics for Plumbers and Pipefitters Unit 2: Formulas
 Unit2a
 MPPUnit2
 Class Discussion: Define the Kinds of Measurements Used and Review of Mathematical Symbols
 Class Discussion: Define the Use of Symbols
 Class Discussion: Relate Geometry to Plumbing
 Class Discussion: Review Math Procedures and Precedence
 CLASS DISCUSSION, Math
 On Campus Pass-off
 Study Resources
 Hybrid Assignments
 Math: Unit 47-48
 MPPUnit3
 Unit4a

Read Mathematics for Plumbers and Pipefitters Unit 5: Review of Angle Measure
 Read Mathematics for Plumbers and Pipefitters Unit 6: Conversion of length Measures
 Math: Unit6
 Pressure Formula
 Class Discussion: Determine How to Find the Square Root of a Number
 Class Discussion: Define an Angle
 Class Discussion: Using Square Roots to Solve Right Triangle Problems
 Class Discussion: Use of Proper Measurements for Each Task
 Class Discussion: Alternate and Corresponding Angles
 Class Discussion: Convert Between Different Dimensional Standards
 On Campus Pass-off
 Study Resources
 Hybrid Assignments
 CLASS DISCUSSION BOARD
 Class activity: Field trip to central davis sewer district plant located at 2200 South Sunset Drive, Kaysville at 6PM
 Grade, Drop, Run
 Math Review
 Math Review Quiz
 On Campus Pass-off I Study Resources
 Hybrid Assignments
 dd IPC SS 4 Quiz - Fixtures, Faucets and Fixture Fittings I
 CLASS DISCUSSION BOARD
 dd IPC SS S Quiz• Fixtures, Faucets and Fixture Fittings II
 On Campus Pass-off I Study Resources
 Hybrid Assignments
 R317-4 Onsite Wastewater System Worksheet
 Competency Test Review
 On Campus Pass-off I Study Resources
 Hybrid Assignments
 Class Discussion: Comp Test Review in Class,
 On Campus Pass-off
 Plumbing 1A Competency Test {1A Final Exam)
 Study Resources
 Hybrid Assignments
 On Campus Pass-off I Competency Test Retake
 Study Resources
 End of Course Survey

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Start: 08-25-2026

End: 12-08-2026

Tu, W

6:00 PM - 9:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jason Parsons — jparsons@stech.edu

Wade Esplin — wesplin@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

You will also be required to complete the course, exams and homework, with 75% or higher accuracy for credit and DOPL licensure points.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation, and student services will be notified.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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